



Lakshya JEE 2.0 (2024)

Probability

DPP-02

- If A and B are independent events such that $P(B) = \frac{2}{7}$, $P(A \cup \bar{B}) = 0.8$, then $P(A) =$
 - 0.1
 - 0.2
 - 0.3
 - 0.4
- The records of a hospital show that 10% of the cases of a certain disease are fatal. If 6 patients are suffering from the disease, then the probability that only three will die is:
 - 1458×10^{-5}
 - 1458×10^{-6}
 - 41×10^{-6}
 - 8748×10^{-5}
- If A and B are two events (such that $P(A) = \frac{3}{4}$ and $P(B) = \frac{5}{8}$), then
 - $P(A \cup B) \geq 3/4$
 - $P(A' \cap B) \leq 1/4$
 - $3/8 \leq P(A \cap B) \leq 5/8$
 - All of the above
- Ten equally spaced points are lying on a circle. If p is the probability that a chosen chord is diameter of the circle, then $18p + 2$ is equal to ____.
- A function is selected at random from all the function defined over a given set A consisting of three distinct elements. The probability that the mapping selected is one to one is
 - $\frac{2}{9}$
 - $\frac{1}{9}$
 - $\frac{3!}{3^3}$
 - $\frac{{}^3P_3}{27}$
- A matrix is chosen at random from the set of all matrices of order 2 with elements 0 or 1 only. The probability that the determinant value of the matrix chosen has the value non-negative is greater than or equals to
 - $\frac{3}{8}$
 - $\frac{13}{16}$
 - $\frac{1}{4}$
 - $\frac{15}{16}$
- The sum of the digits of a seven digit number is 59. If the probability that the number is divisible by 11 is p . then $441p$ is equal to
- If 4-whole numbers taken at random are multiplied together, then find the Probability that the last digit in the product is 1, 3, 7 or 9.
 - $\frac{2}{7}$
 - $\frac{16}{625}$
 - $\frac{3}{17}$
 - $\frac{2}{5}$
- From a group of 10 persons consisting of 5 lawyers, 3 doctors and 2 engineers, four persons are selected at random. The probability that the selection contains at least one of each category is
 - $1/2$
 - $1/3$
 - $2/3$
 - None of these
- If A and B are two events such that $P(A) = 0.3$, $P(B) = 0.25$, $P(A \cap B) = 0.2$, then $P\left(\left(\frac{A^C}{B^C}\right)^C\right)$ is equal to:
 - $\frac{2}{15}$
 - $\frac{11}{15}$
 - $\frac{13}{15}$
 - $\frac{14}{15}$

11. A pack of playing cards was found to contain only 51 cards. If the first 13 cards which are examined are all red, then the probability that the missing card is black, is
- (1) $\frac{1}{3}$ (2) $\frac{2}{3}$
 (3) $\frac{15}{26}$ (4) $\frac{16}{39}$
12. There are two bags, one of which contains 3 black and 4 white balls, while the other contains 4 black and 3 white balls. A die is cast. If the face 1 or 3 turns up a ball is taken out from the first bag and if any other face turns up, a ball is taken from the second bag. The probability of choosing a black ball, is
- (1) $\frac{7}{15}$ (2) $\frac{8}{15}$
 (3) $\frac{10}{21}$ (4) $\frac{11}{21}$
13. There are two groups of subjects, one of which consists of 5 Science subjects and 3 Engineering subjects and the other consists of 3 Science and 5 Engineering subjects. An unbiased die is cast. If number 3 or 5 turns up, a subject from group I is selected, otherwise a subject is selected from group II. The probability that an Engineering subject is selected ultimately, is
- (1) $\frac{7}{13}$ (2) $\frac{9}{17}$
 (3) $\frac{13}{24}$ (4) $\frac{11}{20}$
14. A person is known to speak the truth 4 times out of 5. He throws a die and reports that it is a six. The probability that it is actually a six, is:
- (1) $\frac{1}{3}$ (2) $\frac{2}{9}$
 (3) $\frac{4}{9}$ (4) $\frac{5}{9}$
15. The B.D. whose mean is 3 and variance is 2, is given by ____.
- (1) $B\left(3, \frac{1}{3}\right)$
 (2) $B\left(5, \frac{1}{3}\right)$
 (3) $B\left(6, \frac{1}{3}\right)$
 (4) $B\left(9, \frac{1}{3}\right)$
16. A fair coin is tossed a fixed number of times. If the probability of getting seven heads is equal to that of getting nine heads, the probability of getting two heads is
- (1) $\frac{15}{2^8}$
 (2) $\frac{2}{15}$
 (3) $\frac{15}{2^{13}}$
 (4) none of these
17. A box contains N coins, of which m are fair and the rest are biased. The probability of getting head when a fair coin is tossed is $\frac{1}{2}$, while $\frac{2}{3}$ when a biased coin is tossed. A coin is drawn from the box at random and is tossed twice. The first time it shows head and the second time it shows tail. The probability that the coin drawn is fair, is
- (1) $\frac{5m}{m+8N}$
 (2) $\frac{3m}{m+8N}$
 (3) $\frac{7m}{m+8N}$
 (4) $\frac{9m}{m+8N}$



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (3)
2. (1)
3. (4)
4. (4)
5. (3)
6. (2)
7. (84)
8. (2)
9. (1)

10. (1)
11. (2)
12. (4)
13. (3)
14. (3)
15. (4)
16. (3)
17. (4)



PW Web/App - <https://smart.link/7wwosivoicgd4>

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